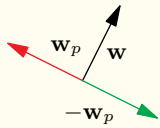
1.1.1

Gitt en vektor $\mathbf{w}$. Vi definerer $\mathbf{w}_p$ som vektoren som

- står vinkelrett på $\mathbf{w}$.
- peker til venstre med hensyn til retningen til $\mathbf{w}$.
- har lik lengde som $\mathbf{w}$.

Hvis $\mathbf{w} = [x, y]$, kan det vises at $\mathbf{w}_p = [-y, x]$, og at $-\mathbf{w}_p$ peker til høyre for $\mathbf{w}$.



Gitt vektorene $\mathbf{u} = [a, b]$, $\mathbf{u}_p = [-b, a]$, $\mathbf{v} = [c, d]$ og $\mathbf{v}_p = [-d, c]$. Vis at

a) $\mathbf{u} \cdot \mathbf{v}_p + \mathbf{u}_p \cdot \mathbf{v} = 0$

b) $(\mathbf{u}_p)_p = -\mathbf{u}$

c) $(\mathbf{u} + \mathbf{v})_p = \mathbf{u}_p + \mathbf{v}_p$

Svar

a) $[a, b] \cdot [-d, c] + [-b, a] \cdot [c, d] = -ad + bc - bd + ad = 0$

b) $(\mathbf{u}_p)_p = ([-b, a])_p = [-a, -b] = -\mathbf{u}$.

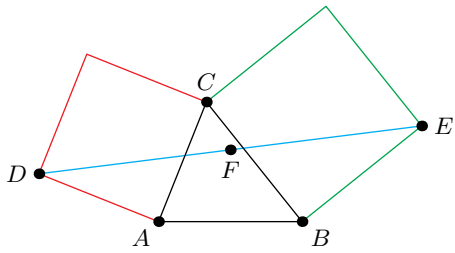
c)
$$\begin{aligned}(\mathbf{u} + \mathbf{v})_p &= ([a + c, b + d])_p \\ &= [-(b + d), a + c] \\ &= [-b, a] + [-d, c] \\ &= \mathbf{u}_p + \mathbf{v}_p\end{aligned}$$

1.1.2

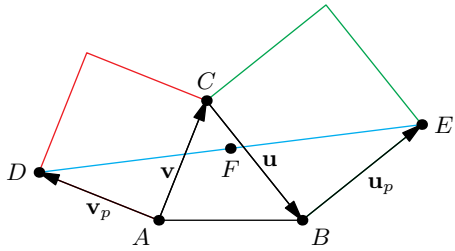
Den røde og den grønne firkanten er begge et kvadrat. Vi setter $\mathbf{v} = \overrightarrow{AC}$ og $\mathbf{u} = \overrightarrow{CB}$. F er midtpunktet på DE.

a) Finn koordinatene til F uttrykt ved $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u}_p$ og $\mathbf{v}_p$.

b) Beskriv den eventuelle forflytningen til F hvis C forskyves i en vilkårlig retning. (Det holder å se på tilfellet hvor C ligger over AB).



Svar



Vi innfører et koordinatsystem med origo i A. Da er $D = \mathbf{v}_p$. Videre er $E = \mathbf{u} + \mathbf{v} + \mathbf{u}_p$. Videre er $\overrightarrow{DE} = -\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p$ Dermed er

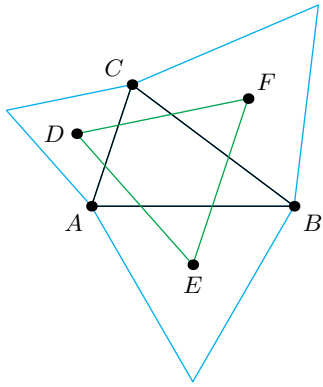
$$F = D + \frac{1}{2}\overrightarrow{DE} = \frac{1}{2}(\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p) = \frac{1}{2}(\mathbf{u} + \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p)$$

$\mathbf{u}+\mathbf{v}$ og $\mathbf{u}_p+\mathbf{v}_p$ er ortogonale og har lik lengde. Så lenge $\mathbf{u}+\mathbf{v}$ ikke endres, vil altså ikke F forflyttes. En forskyvning av C medfører ingen endring av $\mathbf{u} + \mathbf{v}$, og dermed er poisjonen til F uavhengig av posisjonen til C.

1.1.3

De blå linjestykkene inngår i likesidede trekanter på utsiden av $\triangle ABC$. D, E og F er de respektive midtpunktene til de likesidede trekantene.

Vis at $\triangle DEF$ er likesidet.



.

Svar

Vi setter $\overrightarrow{AC} = 2\mathbf{v}$ og $\overrightarrow{AB} = 2\mathbf{u}$. Da er $\overrightarrow{BC} = 2\mathbf{v} - 2\mathbf{u}$. Videre har vi at

$$D = \mathbf{v} + k\mathbf{v}_p$$

$$E = \mathbf{u} - k\mathbf{u}_p$$

$$F = \mathbf{u} + \mathbf{v} - k(\mathbf{v}_p - \mathbf{u}_p)$$

hvor $k = \frac{1}{\sqrt{3}}$. Altså er

$$\overrightarrow{DF} = \mathbf{u} + k(\mathbf{u}_p - 2\mathbf{v}_p)$$
$$\overrightarrow{DE} = \mathbf{u} - \mathbf{v} - k(\mathbf{u}_p + \mathbf{v}_p)$$
$$\overrightarrow{EF} = \mathbf{v} + k(2\mathbf{u}_p - \mathbf{v}_p)$$

Da er

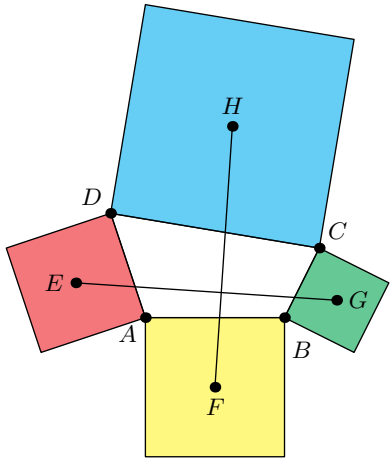
$$\begin{aligned} DF^2 &= \mathbf{u}^2 + 2k\mathbf{u}(\mathbf{u}_p - 2\mathbf{v}_p) + k^2(\mathbf{u}_p - 2\mathbf{v}_p)^2 \\ &= \mathbf{u}^2 - 4k\mathbf{u} \cdot \mathbf{v}_p + k^2(\mathbf{u}_p^2 - 4\mathbf{u}_p \cdot \mathbf{v}_p + 4\mathbf{v}_p^2) \\ &= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p \cdot \mathbf{v}_p) - 4k\mathbf{u} \cdot \mathbf{v}_p \end{aligned}$$

$$
\begin{aligned}
DE^2 &= (\mathbf{u}-\mathbf{v})^2 - 2k(\mathbf{u}-\mathbf{v})(\mathbf{u}_p+\mathbf{v}_p) + k^2(\mathbf{u}_p+\mathbf{v}_p)^2 \\
&= \mathbf{u}^2 - 2\mathbf{u}\cdot\mathbf{v} + \mathbf{v}^2 - 2k(\mathbf{u}\cdot\mathbf{v}_p - \mathbf{u}_p\cdot\mathbf{v}) + k^2(\mathbf{u}_p^2 + 2\mathbf{u}_p\cdot\mathbf{v}_p + \mathbf{v}_p^2) \\
&= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p\cdot\mathbf{v}_p) - 4k\mathbf{u}\cdot\mathbf{v}_p
\end{aligned}
$$

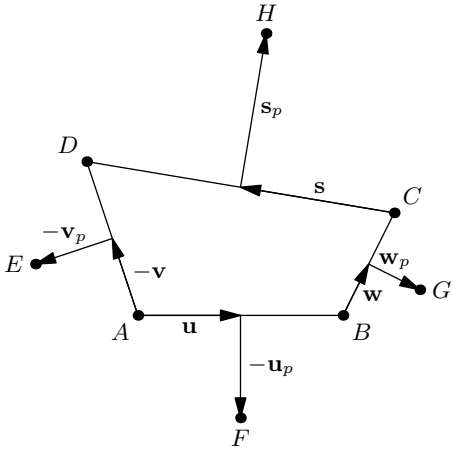
$$
\begin{aligned}
EF^2 &= \mathbf{v}^2 + 2k\mathbf{v}(2\mathbf{u}_p - \mathbf{v}_p) + k^2(2\mathbf{u}_p - \mathbf{v}_p)^2 \\
&= \mathbf{v}^2 + 4k\mathbf{u}_p\cdot\mathbf{v} + k^2(4\mathbf{u}_p - 4\mathbf{u}_p\mathbf{v}_p + \mathbf{v}_p^2) \\
&= \frac{4}{3}(\mathbf{u}^2 + \mathbf{v}^2 - \mathbf{u}_p\cdot\mathbf{v}_p) - 4k\mathbf{u}\cdot\mathbf{v}_p
\end{aligned}
$$

1.1.4

De fargede firkantene er kvadrater. E, F, G og H er de respektive midtpunktene til kvadratene. Vis at $EG = FH$ og $\overrightarrow{EG} \perp \overrightarrow{FH}$.



Svar



Vi innfører et koordinatsystem med origo i A, og setter $\overrightarrow{DA} = 2\mathbf{v}$, $\overrightarrow{AB} = 2\mathbf{u}$, og $\overrightarrow{BC} = 2\mathbf{w}$ (i figuren er $\mathbf{s} = \mathbf{u} + \mathbf{v} + \mathbf{w}$). Da er

$$
\begin{aligned}
E &= -\mathbf{v} - \mathbf{v}_p \\
F &= \mathbf{u} - \mathbf{u}_p \\
G &= 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p \\
H &= \mathbf{u} + \mathbf{w} - \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p
\end{aligned}
$$

Dermed er

$$
\begin{aligned}
\overrightarrow{EG} &= 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p + \mathbf{v} + \mathbf{v}_p = 2\mathbf{u} + \mathbf{w} + \mathbf{v} + \mathbf{v}_p - \mathbf{w}_p \\
\overrightarrow{FH} &= \mathbf{w} - \mathbf{v} + 2\mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p
\end{aligned}
$$

Vi har at

$$\overrightarrow{FH}_p = \mathbf{w}_p - \mathbf{v}_p - 2\mathbf{u} - \mathbf{v} - \mathbf{w} = -\overrightarrow{EG}$$

Dette betyr at $\overrightarrow{FH}$ og $\overrightarrow{EG}$ er ortogonale. Videre er

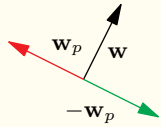
$$FH = |\overrightarrow{FH}_p| = | - \overrightarrow{EG}| = EG$$

1.1.5

Given a vector $\mathbf{w}$. We define $\mathbf{w}_p$ as the vector that

- is perpendicular to $\mathbf{w}$.
- points to the left with respect to the direction of $\mathbf{w}$.
- has the same length as $\mathbf{w}$.

If $\mathbf{w} = [x, y]$, it can be shown that $\mathbf{w}_p = [-y, x]$, and that $-\mathbf{w}_p$ points to the right of $\mathbf{w}$.



Given the vectors $\mathbf{u} = [a, b]$, $\mathbf{u}_p = [-b, a]$, $\mathbf{v} = [c, d]$, and $\mathbf{v}_p = [-d, c]$. Show that

a) $\mathbf{u} \cdot \mathbf{v}_p + \mathbf{u}_p \cdot \mathbf{v} = 0$

b) $(\mathbf{u}_p)_p = -\mathbf{u}$

c) $(\mathbf{u} + \mathbf{v})_p = \mathbf{u}_p + \mathbf{v}_p$

Answer

a) $[a, b] \cdot [-d, c] + [-b, a] \cdot [c, d] = -ad + bc - bd + ad = 0$

b) $(\mathbf{u}_p)_p = ([-b, a])_p = [-a, -b] = -\mathbf{u}$.

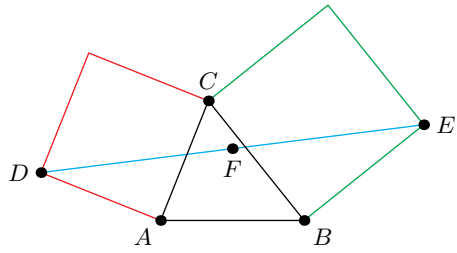
c)
$$\begin{aligned}(\mathbf{u} + \mathbf{v})_p &= ([a + c, b + d])_p \\ &= [-(b + d), a + c] \\ &= [-b, a] + [-d, c] \\ &= \mathbf{u}_p + \mathbf{v}_p\end{aligned}$$

Problem: Bottema's theorem

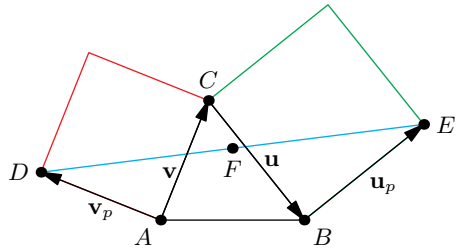
The red and the green quadrilateral are both squares. F is the midpoint of DE. Let $\mathbf{v} = \overrightarrow{AC}$ and $\mathbf{u} = \overrightarrow{CB}$. Given a vector $\mathbf{w}$, we define $\mathbf{w}_p$ as the vector perpendicular to $\mathbf{w}$, pointing to the left with respect to the direction of $\mathbf{w}$, and with equal length as $\mathbf{w}$.

a) Find the coordinates of F expressed in terms of $\mathbf{u}$, $\mathbf{v}$, $\mathbf{u}_p$, and $\mathbf{v}_p$.

b) Describe the possible displacement of F if C is shifted in an arbitrary direction. (It is sufficient to consider the case where C is above AB).



Solution



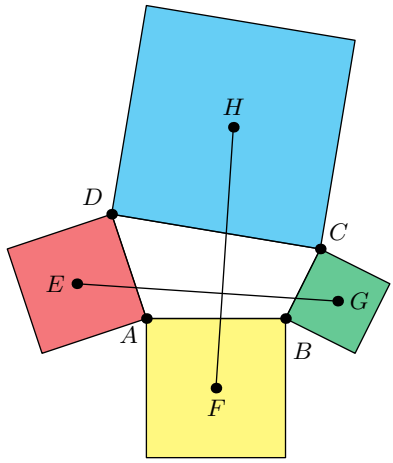
We introduce a coordinate system with the origin at A. Then $D = \mathbf{v}_p$. Furthermore, $E = \mathbf{u} + \mathbf{v} + \mathbf{u}_p$. Furthermore, $\overrightarrow{DE} = -\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p$. Thus,

$$F = D + \frac{1}{2}\overrightarrow{DE} = \frac{1}{2}(\mathbf{v}_p + \mathbf{v} + \mathbf{u} + \mathbf{u}_p) = \frac{1}{2}(\mathbf{u} + \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p)$$

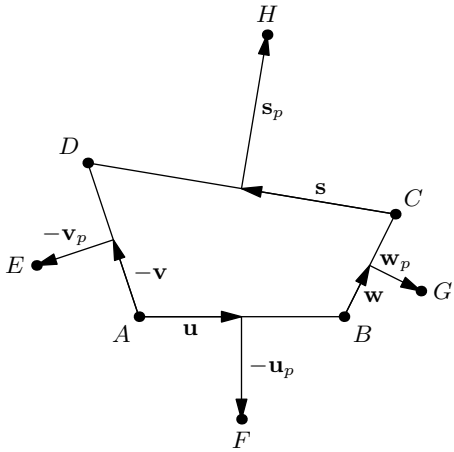
$\mathbf{u} + \mathbf{v}$ and $\mathbf{u}_p + \mathbf{v}_p$ are orthogonal and have the same length. As long as $\mathbf{u} + \mathbf{v}$ does not change, F will not be displaced. A displacement of C does not change $\mathbf{u} + \mathbf{v}$, and thus the position of F is independent of the position of C.

Problem: van Aubel's theorem

The colored quadrilaterals are squares. E, F, G, and H are the respective midpoints of the squares. Show that $EG = FH$ and $\overrightarrow{EG} \perp \overrightarrow{FH}$.



Answer



We introduce a coordinate system with the origin at A, and set $\overrightarrow{DA} = 2\mathbf{v}$, $\overrightarrow{AB} = 2\mathbf{u}$, and $\overrightarrow{BC} = 2\mathbf{w}$ (in the figure, $\mathbf{s} = \mathbf{u} + \mathbf{v} + \mathbf{w}$). Then,

$$
\begin{aligned}
E &= -\mathbf{v} - \mathbf{v}_p \\
F &= \mathbf{u} - \mathbf{u}_p \\
G &= 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p \\
H &= \mathbf{u} + \mathbf{w} - \mathbf{v} + \mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p
\end{aligned}
$$

Thus,

$$\overrightarrow{EG} = 2\mathbf{u} + \mathbf{w} - \mathbf{w}_p + \mathbf{v} + \mathbf{v}_p = 2\mathbf{u} + \mathbf{w} + \mathbf{v} + \mathbf{v}_p - \mathbf{w}_p$$
$$\overrightarrow{FH} = \mathbf{w} - \mathbf{v} + 2\mathbf{u}_p + \mathbf{v}_p + \mathbf{w}_p$$

We have

$$\overrightarrow{FH}_p = \mathbf{w}_p - \mathbf{v}_p - 2\mathbf{u} - \mathbf{v} - \mathbf{w} = -\overrightarrow{EG}$$

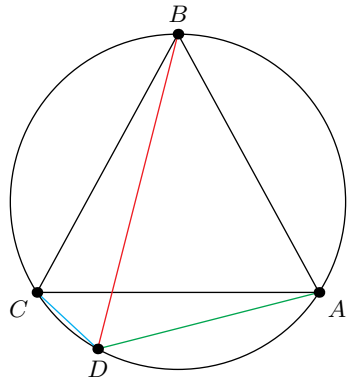
This means that $\overrightarrow{FH}$ and $\overrightarrow{EG}$ are orthogonal. Furthermore,

$$FH = |\overrightarrow{FH}_p| = | - \overrightarrow{EG}| = EG$$

A, B, C og D ligger på sirkelen, og $\angle ACB = \angle BAC = v$.

Vis at

$$BD = \frac{CD + DA}{2\cos v}$$



Svar

$\angle BDC$ og $\angle ADB$ spenner over samme bue som henholdsvis $\angle ACB$ og $\angle BAC$. Alle disse er periferivinkler, og da $\angle ACB = \angle BAC$, har de alle like størrelse. Videre følger det at $BC = AB$. Av cosinussetingen har vi da at

$$\begin{aligned} CD^2 + BD^2 - 2CD \cdot BD\cos v &= DA^2 + BD^2 - 2DA \cdot BD\cos v \\ CD^2 - DA^2 &= 2BD\cos v(CD - DA) \\ (CD + DA)(CD - DA) &= 2BD\cos v(CD - DA) \end{aligned}$$

Antatt at $CD - DA \neq 0$, er

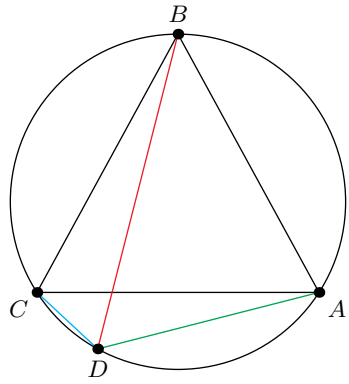
$$BD = \frac{CD + DA}{2\cos v}$$

I tilfellet hvor $CD = DA$, er BD en diameter i sirkelen. Da har vi av Thales' setning omvendt at $\angle DCB = 90°$. Altså er

$$BC = \frac{CD}{\cos v} = \frac{CD + DA}{2\cos v}$$

A, B, C and D lies on the circle, and $\angle ACB = \angle BAC = v$. Show that

$$BD = \frac{CD + DA}{2\cos v}$$

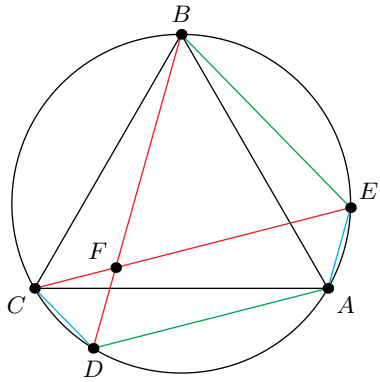


$\triangle ABC$ er likesidet, og $\triangle ACD \cong \triangle BAE$.

a) Avgjør hvilke av disse vinklene som har lik verdi. Bestem vinkelverdien om det lar seg gjøre.

$$\angle BDC, \angle ADB, \angle CFD, \angle ACE, \angle BEC, \angle EFB \text{ og } \angle CAD$$

b) Vis at $BD = CD + DA$.



Svar

a) $\angle BDC = \angle BEC = \angle BAC = 60°$ fordi de er periferivinkler som spenner over samme bue. Tlsvarende kan vi finne at $\angle ADB = 60°$. $\angle CAD = \angle ACE$ fordi de er periferivinkler som spenner over sirkelsegmenter med lik lengde. Av vinkelsummen til $\angle ADC$ er $\angle DCA = 60 - \angle CAD$. Dermed er

$$\angle DCF = \angle DCA + ACE = 60°$$

Altså er $\triangle CDF$ likesidet. $\angle EFB = \angle CFD$ fordi de er toppvinkler, og dermed er også $\triangle FEB$ likesidet. Følgelig har vi at

$$\angle BDC = \angle ADB = \angle CFD = \angle BEC = \angle EFB = 60°$$

og

$$\angle CAD = \angle ACE$$

b) Da $\triangle CDF$ og $\triangle FEB$ er likesidet, har vi at

$$BD = BF + FD = BE + CD = AD + CD$$

The Quatratic Function

Problem 1

Given the function $f(x) = ax^2 + bx + c$. Show that the graph of f is symmetric about the line $x = -\frac{b}{2a}$.

Answer

For the graph of f to be symmetric about the line $x = -\frac{b}{2a}$, for any number k, we must have

$$f\left(k - \frac{b}{2a}\right) = f\left(-k - \frac{b}{2a}\right) \tag{1.1}$$

For a number d, we have

$$d - \frac{b}{2a} = \frac{2ad - b}{2a}$$

Furthermore,

$$\begin{aligned} f\left(d - \frac{b}{2a}\right) &= a\left(\frac{2ad - b}{2a}\right)^2 + b\left(\frac{2ad - b}{2a}\right) + c \\ &= \frac{4a^2d^2 - 4abd + b^2}{4a} + \frac{2abd - b^2}{2a} + c \\ &= \frac{4a^2d^2 - b^2}{4a} + c \end{aligned}$$

This means that regardless of whether $d = k$ or $d = -k$, the expression above will be the same, and thus (1.1) holds.

Problem 2

Given the function $f(x) = ax^2 + bx + c$. In Problem 1, we showed that f is symmetric about the line $x = \frac{-b}{2a}$. Furthermore, it can be shown that for all numbers k,

$$f\left(\pm k - \frac{b}{2a}\right) = \frac{4a^2k^2 - b^2}{4a} + c$$

Use this to derive the *abc*-formula.

Answer

If $f(x) = ax^2 + bx + c = 0$, then for a number k, we have

$$f\left(\pm k - \frac{b}{2a}\right) = \frac{4a^2k^2 - b^2}{4a} + c = 0.$$

Solving this equation for k, we obtain

$$k = \pm\frac{\sqrt{b^2 - 4ac}}{2a}.$$

This means that

$$f\left(\pm\frac{\sqrt{b^2 - 4ac}}{2a} - \frac{b}{2a}\right) = 0.$$

Problem 3

Given the function $f(x) = ax^2 + bx + c$.

a) Find an expression for the extremal value.

b) Explain why

- a change in c will result in a vertical shift of the graph of f.
- a change in a or b will result in both a horizontal and a vertical shift of the graph of f.

Answer

In this solution, we assume the function has a minimum point. The case where it has a maximum point is entirely analogous.

a) Since $x = -\frac{b}{2a}$ is the extremal point of f, the extremal value f_e is given by

$$f_{\mathrm{e}} = f\left(-\frac{b}{2a}\right) = a\left(-\frac{b}{2a}\right)^2 + b\left(-\frac{b}{2a}\right) + c = -\frac{b^2}{4a} + c$$

b) Since the extremal point is expressed in terms of a and b, a change in c will not affect the horizontal position of the minimum point. However, c is part of the expression for f_e, so a change in c will alter the vertical position of the minimum point.

Both a and b are part of the expressions for the extremal point and f_e. Thus, a change in a or b will shift the minimum point both horizontally and vertically.

A shift in the minimum point necessarily corresponds to a shift in the graph of f.

Problem 4

Given the function $f(x) = ax^2 + bx + c$, where[1] $a > 0$. To describe the shape of the graph of f, we do the following[2]:

- Let $g(x)$ be the function that describes the difference between $f(x)$ and the minimum value of f.
- Let $h(g)$ be the horizontal distance between $g(x)$ and the line $x = -\frac{b}{2a}$.

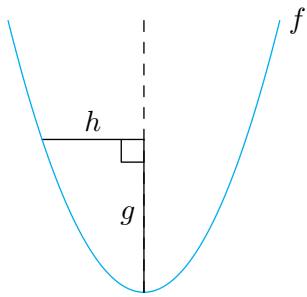
If two quadratic functions have identical expressions for $h(g)$, we can say that the functions have the same shape.

a) Show that

$$h(g) = \sqrt{\frac{g}{a}}$$

b) Refer back to Problem 3, and create a summary of how a change in a, b, or c might affect

- the horizontal shift of the graph,
- the vertical shift of the graph,
- the shape of the graph.



[1] The case where $a < 0$ is entirely analogous.

[2] The case where f has a maximum point instead of a minimum point is entirely identical.

Answer

a) In Problem 3, we found that the extremal value f_e of f is

$$f_e = -\frac{b^2}{4a} + c$$

Therefore,

$$g(x) = f(x) - f_e = ax^2 + bx + c - \left(-\frac{b^2}{4a} + c\right) = ax^2 + bx + \frac{b^2}{4a}$$

Solving this equation for x, we get from the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4a\left(\frac{b^2}{4a} - g\right)}}{2a} = \frac{-b \pm 2\sqrt{ag}}{2a}$$

Let $p(g) = x$ be the inverse function of g for $x \in [-\infty, -\frac{b}{2a}]$. Then,

$$p(g) = \frac{-b - 2\sqrt{ag}}{2a}$$

Furthermore,

$$h(g) = p(g) - \left(-\frac{b}{2a}\right) = \sqrt{\frac{g}{a}}$$

b) A change in c vertically shifts the graph but does not change its shape.

A change in b shifts the graph both horizontally and vertically but does not change its shape.

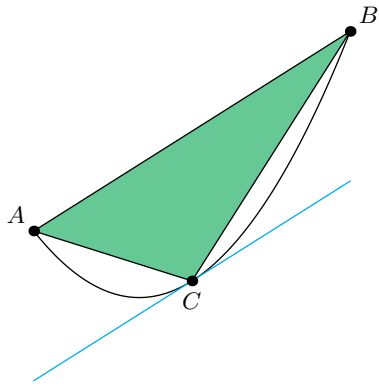
A change in a shifts the graph both horizontally and vertically and changes the shape of the graph.

Problem 5

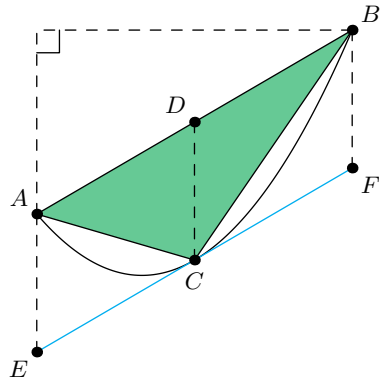
Given the function $f(x) = ax^2+bx+c$, and the points $A = (p, f(p))$ and $B = (q, f(q))$. The tangent to f at the point $C = (t, f(t))$ is parallel to the segment AB.

a) Show that $t = \frac{1}{2}(p + q)$.

b) Show that the area of $\triangle ABC$ is $\frac{1}{8}(q - p)^3$.



Answer



a) We have

$$A = (p, ap^2 + bp + c) \qquad \text{and} \qquad B = (q, aq^2 + bq + c)$$

Let $g(x)$ be the function describing the line through AB. Then,

$$\begin{aligned} g'(x) &= \frac{f(q) - f(p)}{q - p} \\ &= \frac{aq^2 + bq + c - (ap^2 + bp + c)}{q - p} \\ &= \frac{a(q^2 - p^2) + b(q - p)}{q - p} \\ &= \frac{a(q + p)(q - p) + b(q - p)}{q - p} \\ &= a(q + p) + b \end{aligned}$$

Let $C = (t, f(t))$. Then we have

$$\begin{aligned} f'(t) &= a(q + p) + b \\ 2at + b &= a(q + p) + b \\ t &= \frac{1}{2}(q + p) \end{aligned}$$

b) Let D be the midpoint of AB. Then,

$$D = \left(t, \frac{1}{2}(f(q) + f(p))\right)$$

Furthermore,

$$\begin{aligned}
DC &= \frac{1}{2}\left(f(q)+f(p)\right)-f(t)\\
&= \frac{1}{2}\Big(a(p^2+q^2)+b(p+q)+2c\Big)-a\left(\frac{1}{2}(p+q)\right)^2+b\left(\frac{1}{2}(p+q)\right)+c\\
&= \frac{a}{2}(p^2+q^2)-\frac{a}{4}(p+q)^2\\
&= \frac{a}{4}(2p^2+2q^2-p^2-q^2-2pq)\\
&= \frac{a}{4}(q-p)^2
\end{aligned}$$

We place E and F such that $\square AEFB$ is a parallelogram. With AE as the base, the height of $\square AEFB$ is $q-p$. Since $AE=DC$, we have

$$A_{\triangle ABC}=\frac{1}{2}A_{\square AEFB}=\frac{1}{2}\cdot\frac{a}{4}(q-p)^2(q-p)=\frac{a}{8}(q-p)^3$$

Problem 6

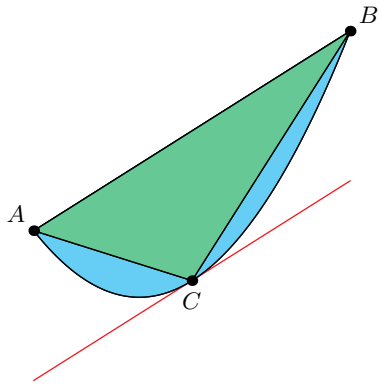
Given the function $f(x) = ax^2+bx+c$ and the points $A = (p, f(p))$ og $B = (q, f(q))$.

a) Let I be the area bounded by the graph of f and the line segment AB. Determine which of a, b, c, p and q we can expect to be included in the expression for I.

b) Find the expression for I.

c) Find $\frac{I}{A_{\triangle ABC}}$.

d) Find $\frac{A_{\triangle ABC}}{A_{\text{blue area}}}$.



Answer

a) We expect p and q to appear in the expression for I since these determine the positions of A and B. As long as the shape of f remains the same, the bounded area will not change. Since only a affects the shape of f, a must also appear in the expression for I, but not b and c.

b) Let $g(x)$ be the function describing the line through AB. Then

$$\begin{aligned} g(x) &= (a(p+q)+b)(x-p)+f(p) \\ &= (a(p+q)+b)x-(a(p+q)+b)p+ap^2+bp+c \\ &= (a(p+q)+b)x-apq+c \end{aligned}$$

We have that

$$g-f = ax(p+q)-apq-ax^2$$

Therefore,

$$\begin{aligned} I &= \int_p^q (g-f)\,dx \\ &= \left[\frac{ax^2}{2}(p+q)-apqx-\frac{1}{3}ax^3\right]_p^q \\ &= \frac{aq^2}{2}(p+q)-apq^2-\frac{1}{3}aq^3-\left(\frac{ap^2}{2}(p+q)-ap^2q-\frac{1}{3}ap^3\right) \\ &= \frac{a}{2}(p+q)(q^2-p^2)-apq(q-p)-\frac{a}{3}(q^3-p^3) \\ &= \frac{a}{2}(q-p)((p+q)^2-2pq)-\frac{a}{3}(q^3-p^3) \\ &= \frac{a}{2}(q-p)(p^2+q^2)-\frac{a}{3}(q^3-p^3) \\ &= \frac{a}{2}(q-p)(p^2+q^2)-\frac{a}{3}(q-p)(p^2+q^2+pq) \\ &= \frac{a}{6}(q-p)(3p^2+q^2-2p^2-2q^2-2pq) \\ &= \frac{a}{6}(q-p)^3 \end{aligned}$$

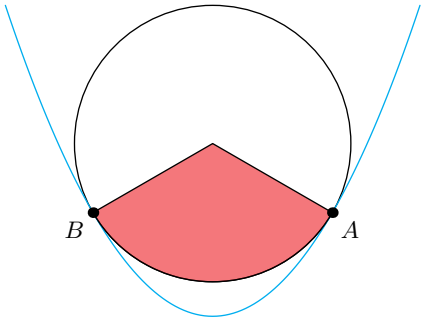
c) Since $I = \frac{a}{6}(q-p)^3$ and $A_{\triangle ABC} = \frac{a}{8}(q-p)^3$, we have

$$\frac{I}{A_{\triangle ABC}} = \frac{4}{3}$$

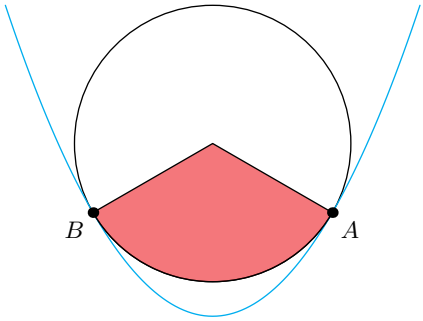
d) We have that

$$A_{\text{blue area}} + A_{\triangle ABC} = \frac{4}{3} A_{\triangle ABC}$$
$$\frac{A_{\triangle ABC}}{A_{\text{blue area}}} = 3$$

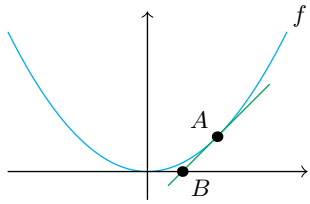
Grafen til $f = x^2 + \frac{3}{4}$ tangerer sirkelen i A og B. $AB = \sqrt{3}$.
Finn arealet til den røde sektoren.



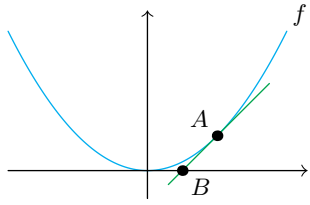
The graph of $f = x^2 + \frac{3}{4}$ are tangent to the circle at A and B.
$AB = \sqrt{3}$.
Find the area of the red sector.



Gitt funksjonen $f(x) = ax^2$. Tangenten til f i punktet A skjærer x-aksen i punktet B. Vis at hvis $A = (c, f'(c))$, så er $B = \left(\frac{c}{2}, 0\right)$.



Given the function $f(x) = ax^2$. The tangent to f at point A intersects the x-axis at point B. Show that if $A = (c, f(c))$, then $B = \left(\frac{c}{2}, 0\right)$.



Gitt funksjonen $f(x) = x^2$. Det finnes et **brennpunkt** $F = (0, p)$ og en **styrelinje** gitt ved $s(x) = -p$ slik at $AF = AB$, hvor A ligger på grafen til f, $AF \perp AB$, og $p \neq 0$.

a) Finn verdien til p.

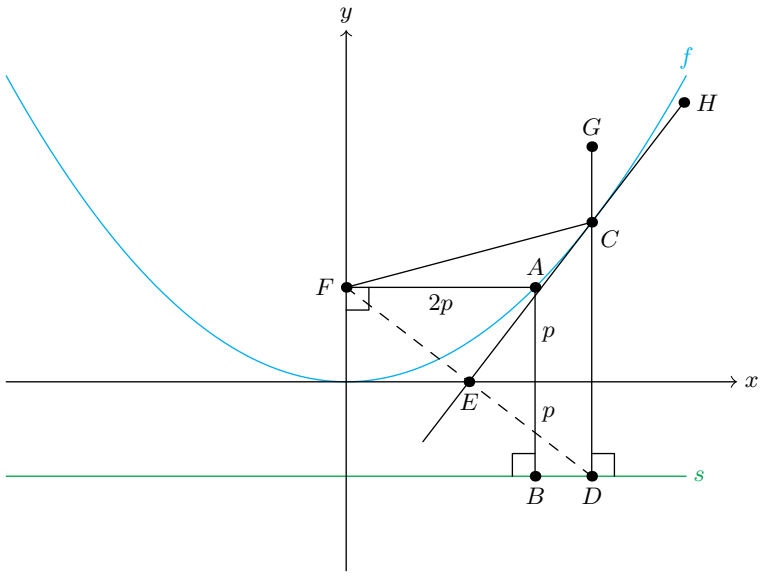
Vi setter $C = (c, f(c))$.

b) Finn lengden til FC uttrykt ved c.

c) Vis at $FC = CD$.

d) E er skjæringspunktet til tangenten til f i C og x-aksen. Finn x-koordinaten til E, og vis at E ligger på FD.

e) Vis at $\angle HCG = \angle FCE$.



Given the function $f(x) = x^2$. There exists a **focus** $F = (0, p)$ and a **directrix** given by $s(x) = -p$ such that $AF = AB$, where A lies on the graph of f, $AF \perp AB$, and $p \neq 0$.

a) Find the value of p.

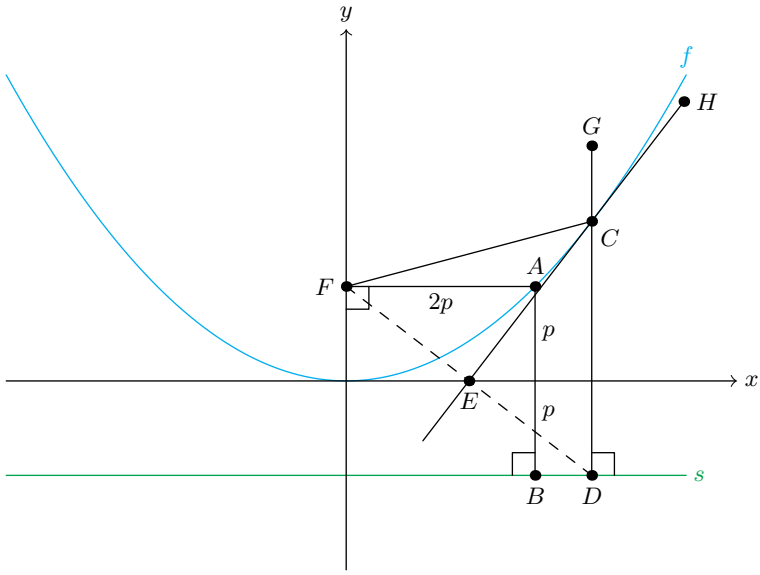
Let $C = (c, f(c))$.

b) Find the length of FC expressed in terms of c.

c) Show that $FC = CD$.

d) E is the intersection point of the tangent to f at C and the x-axis. Find the x-coordinate of E, and show that E lies on FD.

e) Show that $\angle HCG = \angle FCE$.



Given a logistic function

Given a logistic function

$$f(x) = \frac{a}{1 + Ce^{-bx}}$$

where $a, b, C > 0$.

a) Show that $f' = \frac{b}{a} f \cdot (a - f)$.

b) Let d be the number such that $f''(d) = 0$. Find d.

c) Show that $f(d + x) + f(d - x) = a$.

d) Let $h_0 = f(d) - f(d - x)$ and $h_1 = f(d + x) - f(d)$. Show that $h_0 = h_1$. Provide a geometric interpretation of this identity.

e) Show that $f''(d + x) = -f''(d - x)$. What point does this identity suggest that $x = d$ is for f? What point is $(d, f(d))$ for f?